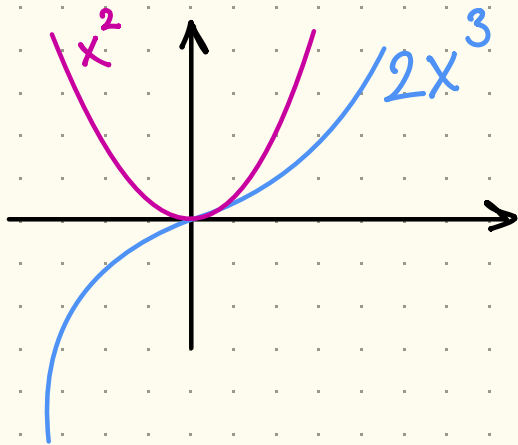


LIMITI (ES)

es) $\lim_{x \rightarrow -\infty} 2x^3 + x^2 - 3?$



x	x^2	x^3
10	10^2	10^3
100	10^4	10^6
1000	10^6	10^9

$$\cdot \lim_{x \rightarrow -\infty} 2x^3 + x^2 - 3$$

$$= \lim_{x \rightarrow -\infty} x^3 \left(2 + \frac{1}{x} - \frac{3}{x^3} \right) = -\infty$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $-\infty$ 2 0 0
└──────────┘
↓
2

es) $\lim_{x \rightarrow +\infty} \frac{\arctan x}{x - 3x^2}$

$$x - 3x^2 = x(1 - 3x) \rightarrow -\infty$$

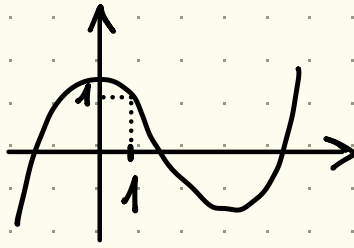
POSSO ANCHE
RACCOLGERE PER x^2

$$= \underbrace{x^2}_{-\infty} \left(\underbrace{\frac{1}{x} - 3}_{3} \right) \rightarrow -\infty$$

es) $\lim_{x \rightarrow 1} \frac{\sin x}{x-1}$ → SI ANALIZZANDO NUM. E DEN. SINGOLARMENTE.

N: $\sin 1$

D: $1-1=0$



• $\lim_{x \rightarrow 1^-} \frac{\sin x}{x-1} = -\infty$
Annotations: $\sin 1 > 0$ (red arrow pointing to the numerator), 0^- (pink arrow pointing to the denominator).

• $\lim_{x \rightarrow 1^+} \frac{\sin x}{x-1} = +\infty$
Annotations: $\sin 1$ (red arrow pointing to the numerator), 0^+ (pink arrow pointing to the denominator).

QUINDI $\nexists \lim_{x \rightarrow 1} \frac{\sin x}{x-1}$

es) $\lim_{x \rightarrow +\infty} \frac{x^2 + 3x}{x - 3x^2}$

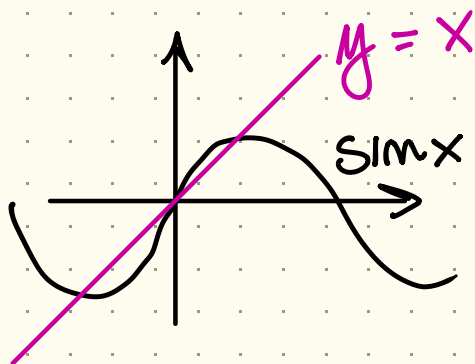
• $\lim_{x \rightarrow +\infty} \frac{x^2 + 3x}{x - 3x^2}$
Annotations: $+\infty$ (red arrow pointing to the numerator), $-\infty$ (pink arrow pointing to the denominator).

• $\frac{x^2 + 3x}{x - 3x^2} = \frac{x^2(1 + 3/x)}{x^2(1/x - 3)} = \frac{1 + 3/x \rightarrow 1}{1/x - 3 \rightarrow -3}$

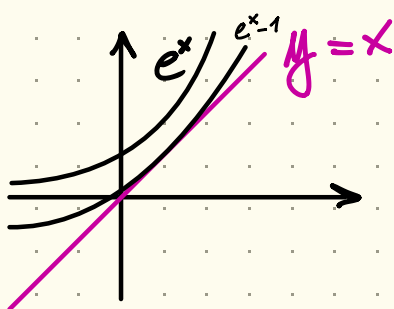
$\downarrow x \rightarrow -\infty$
 $-\frac{1}{3}$

• FORMA INDETERMINATA / LIMITI NOTEVOLI

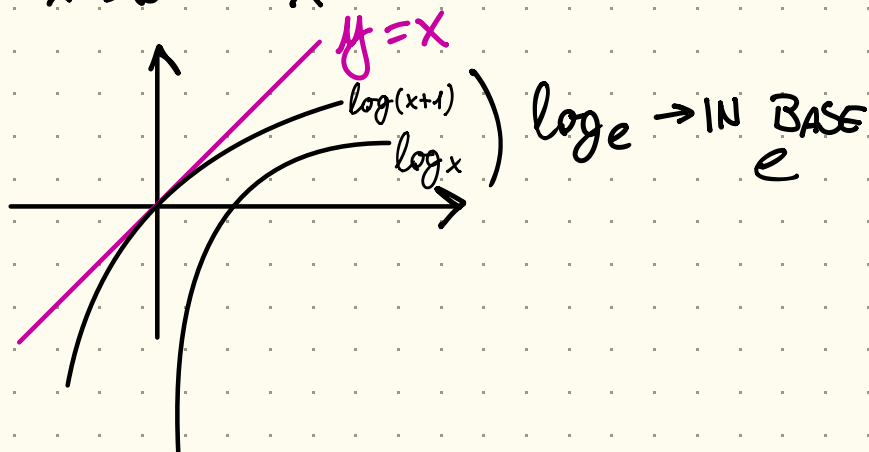
$$\bullet \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$



$$\bullet \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$



$$\bullet \lim_{x \rightarrow 0} \frac{\log(x+1)}{x} = 1$$



$$\cdot \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\cdot \lim_{x \rightarrow 0} \frac{x - \cos x}{x^3} = \frac{1}{6}$$

$$\cdot \lim_{x \rightarrow +\infty} \frac{a^x}{x^n} = +\infty \quad \begin{matrix} a > 1 \\ n \in \mathbb{N} \end{matrix}$$

$$\cdot \lim_{x \rightarrow +\infty} \frac{x^n}{\log x} = +\infty$$

$$\cdot \lim_{x \rightarrow -\infty \rightarrow \pm\infty \rightarrow 0} x^n e^x = 0$$

$$\cdot \lim_{x \rightarrow 0} x \log x = 0$$

$$\cdot \lim_{x \rightarrow +\infty} \log x - x = -\infty$$

• LIMITE FUNZIONE COMPOSTA

$$\cdot \lim_{x \rightarrow x_0} g(f(x)) = \lim_{y \rightarrow y_0} g(y)$$

$$\cdot \lim_{x \rightarrow x_0} f(x) = y_0$$

PER LA COMPOSTA, SVOLGO L'INTERNA E POI L'ESTERNA

$$\cdot \lim_{x \rightarrow 1} \frac{\sin(x-1)}{x-1} = \lim_{y \rightarrow y_0} \frac{\sin y}{y} = 1$$

$$g(y) = \frac{\sin y}{y} \quad y = \underbrace{x-1}_0$$

$$f(x) = x-1$$

$$\lim_{x \rightarrow 1} x - 1 = 0 = y_0$$